



CLIMATE CHANGE AND GENDER ACTION PLAN OF MALI – DRAFT 1

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The National Climate Change and Gender Action Plan (NCCGAP) was jointly conceptualized and developed by staff and consultants from various institutions including AfricaRice, Institute for Rural Economics (IER), National Meteorological Agency (Mali-Meteo), Niger Office (Office du Niger), Syngenta Foundation for Sustainable Agriculture, Rural Polytechnic Institute of Training and Applied Research of Katibougou (IPRA-IFRA Katibokou), Universities of Segou and Bamako, Ministry of Agriculture, Livestock and Fisheries, Rural Development Offices of Barguineda, Segou, and Selingue, Association of Environment and Sustainable Development (AEDD), GRADECOM NGO, and AMACIC NGO.

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ACRONYMS

AEDD	National Agency for Sustainable Development
AHOSCC	Committee of African Heads of States on Climate Change
CAADP	Detailed Program for the Development of Agriculture in Africa
CBD	Convention on Biological Diversity
CBOs	Community-based Organization(s)
CEDAW	Convention on the Elimination of All Forms of Discrimination Against Women
CSA	Climate Smart Agriculture
CSO	Civil Society Organization(s)
DCC	Direction du Changement Climatique,
DGD	Direction du Genre et de Développement
FCF	National Adaptation Program of Action Climate change Fund
GHG	Greenhouse gas
GII	Gender Inequality Index
HDI	Human Development Index
IPCC	Intergovernmental Panel on Climate Change
IUCN	International Union for the Conservation of Nature's
ITCZ	Inter-Tropical Convergence Zone
IWRM	Integrated Water Resources Management
MAEFR	Ministère de l'Agriculture, de l'Élevage et de la Pêche,
MOA	Ministry of Agriculture
NAP	National Adaptation Plan
NCAGF	National Climate Change Gender Action Plan
NGO	Non-governmental Organization(s)
ONAGREF	National Livestock Office
PNISA	Agricultural Sector Investment Plan
REDD	Reducing Emissions from Degradation and Deforestation
SGBV.	Sexual and gender-based violence
UNCCC	United Nations Convention on Climate Change
UNDP	United Nations Development Program
UNECA	United Nations Economic Commission for Africa
UNFCCC	United Nations Convention on Climate
WEDO	Women's Environment and Development Organization
WPCCAA	Work Program on Climate Change Action in Africa

EXECUTIVE SUMMARY

Mali is a landlocked, Sahelian country of West Africa. The Climate Change Vulnerability Index produced by ND-GAIN in 2016 ranked Mali as the 166th out of 181 most at climate change affected-risk countries in the world. It is the 9th most vulnerable country and the 43rd least ready country to address climate change effects. Mali is home to the inner delta of the Niger River that runs through nine countries in the region. The country is influenced by the Inter-Tropical Convergence Zone (ITCZ) which is, creating winds from Ocean and the winds from the Sahara region that are more dusty and warmer. The country is already experiencing the climate change impacts of droughts, flood, storms, strong winds, and increased temperature variability. These events have a devastating impact on crop yields, livestock production, and food security. Available facts show that Mali is among those countries extremely vulnerable to future climate vagaries, when climate change is expectedly going to increase local temperatures, the variability of rainfall, and the magnitude of extreme weather events. Gender relations are central in climate change discourse in Mali as men, women and youths play distinctly disproportionate roles in agriculture in Mali. Women specifically account for over 50% of agricultural labor, and are responsible for the majority of food production. Women, men and youth face varying levels of challenges in the agricultural sector, in accessing key production resource factors including land, water, and credit. Gender inequality defied prior progressive global, regional, and national commitments to protect and promote women's rights. The Agriculture and Climate Change Gender Action Plan for Mali aims to address the gender dimensions of climate change in the agricultural sector. The plan has four key objectives, which include:-

- (i) increase women and youth's access to land, water, and credit.
- (ii) improve knowledge and skill of the various gender groups (women, men, and youths) in climate-smart agriculture with a determined effort to narrow down possible existing gender gaps among the groups.
- (iii) strengthen women's participation in decision-making and policy-making related to climate change.
- (iv) promote the use of gender-sensitive indicators to monitor and evaluate the plan's implementation.

The plan is expected to have a number of benefits, including:

- Increased food security and nutrition for women and their families.
- Improved livelihoods for men, women, and youths.
- Increased resilience to climate change.
- Increased gender equality in the agricultural sector.

The plan will be implemented by a number of government agencies, non-governmental organizations, and development partners. The plan is expected to be fully implemented by 2030.

Key Messages

Concisely, CCGAP has 4 key messages to pass on. I) It re-emphasizes the centrality of climate change in the core of agricultural development discourse showing that Climate change is a major threat to the agricultural sector in Mali. li) It highlights the indispensability of women's role as women play a disproportionately large role in agriculture in Mali. lii) The CCGAP for Mali aims to address the gender dimensions of climate change in the agricultural sector, and Iv) to effectively communicate the principles and goals necessary for integrating gender considerations into climate change strategies and fostering sustainable and inclusive outcomes.

Keywords. Gender, climate change, action plan, investment, inclusive development

SECTION 1: INTRODUCTION

1.1. Background of the study

Mali is one of the hardest climate changes hit West African countries where climate change has come to stay imposing a significant threat to the environment, economy, and society. Projections by the International Panel on Climate Change (IPCC) and downscaled data suggest mean annual temperature in Mali will increase by 1.2 to 3.6°C by 2060 and 1.8 to 5.9°C by 2090 (World Bank, 2018). The increasing frequency in the occurrence of droughts and floods is a common challenge and is projected to intensify under climate change (World Bank, 2018; FAO, 2012). The Southern regions crucial for agriculture, are experiencing land degradation and expansion of the Sahara Desert due to changing rainfall patterns (World Bank, 2018). There is a loss of biodiversity and ecosystem services due to changes in temperature, precipitation, deforestation and agricultural land degradation. Mali is implementing adaptation actions to reduce vulnerability and increase resilience to climate change. The effects of climate change vary across individual groups, localities, regions, and nations and are disproportionate among vulnerable and socio-economic groups. Socio-economic vulnerability of Mali is high. Over 50% of the country GDP relies on agriculture. Climate change is projected to exacerbate food insecurity, potentially pushing an additional 1.3 million people in rural areas below the poverty line by 2050 (FAO, 2012). Climate change effects have imposed real social challenges on the health and livelihood of the people and are inducing stress-migration on the Malian farmers. World Bank (2018) reports increasing heat stress, changes in vector-borne diseases, and malnutrition due to food insecurity are expected to negatively impact public health. Drought, floods, and land degradation threaten the livelihoods of millions of Malians particularly those dependent on agriculture and pastoralism. Climate-induced displacement and migration are likely to increase and put further strain on communities and resources.

The intersection of climate change and gender in Mali presents a complex and crucial issue. Mali's predominantly rural population relies heavily on rain-fed agriculture and natural resources making it highly vulnerable to the impacts of climate change. These impacts often disproportionately affect women and girls due to existing gender inequality. Mali is ranked 155th out of the 170 countries included in the 2021 UNDP Gender Inequality Index (Turkey, 2023). Women and girls in Mali are critically vulnerable when considering land degradation and gender-based violence (UN Women, 2020), food insecurity and water scarcity (World Bank 2018) and, health risk (WHO, 2016) impacts. Climate change stressors exert pressure on scarce agricultural land and water leading to changes in traditional transhumance patterns, greatly increasing pressure on limited forage resources and causing conflicts. Herdsmen drift southwards towards the agricultural zone in search of water and vegetation bringing in violent conflicts involving killings and raping of women and girls. Stress induced out-migration of men leave women with huge responsibilities to cater for households and produce food for the entire nation.

Women are predominant in the labour force. Yet, they take limited advantage of their engagement as the gap between male-female labor force participation rates in Mali ranks second highest in the WAEMU (Turkey, 2023). This suggests that there is scope for significant improvement with potentially large economic benefits. Increasing female participation in public roles offers wide-ranging potential benefits. Dudouet and Schädel, (2021), reveals that participation by women in peace negotiations and constitutional reform processes increased the likelihood that agreements would be reached and implemented. UN Women, (2015) found the probability of a peace agreement lasting at least 15 years to be 35 percent higher when women participate. In the literature compiled by Turkey, (2023), Mali ranked 14th out of almost 180 countries in terms of fragility. Akanbi and others, (2021) reported that sustained policy action is required to overcome the considerable hurdles to the exiting fragility trap. Development in the economic processes that support inclusive growth, can be especially beneficial. This highlights the centrality of policies that favor gender mainstreaming in agricultural system development especially in the urgent issue of tackling treat of climate change.

The limited participation of women and girls in decision-making processes and labor markets deepen inequalities and often prevent them from fully contributing to climate-related planning, policy-making, implementation, monitoring and evaluation. Invariably, climate change has come to stay and women and girls will continue to bear a disproportionate share of the burden of climate change impacts on agriculture as long as inequality is not tackled.

Discourses on these forms of gender inequality have been a source of global debate as reflected in various international conferences targeted towards promoting the status of women. Consequently, different platforms and agencies sponsored by state and non-state actors are set up in Mali, to promote the inclusion of women in all sectors. This particular action plan places emphasis on gender mainstreaming in Mali's agricultural sector (agriculture and livestock production), so that both men and women's unique needs are reflected in policy formulation and implementation in line with FAO, (2019).

1.2. Climate change and gender action plan (CCGAP) process

The CCGAPs refers to the development of gender-responsive national climate change strategies, and marks a key achievement of the International Union for the Conservation of Nature's (IUCN) in Partnership with governments and civil societies. They represent a country's intention to empower and respond to the needs of the voiceless - often invisible sub-population in the context of climate change. They link national and global policy to form strong synergy in, communicating the relevance of women involvement in the national and international community projects/programs. Most importantly, CCGAPs have the potential to enhance the effectiveness and efficiency of climate change and socioeconomic development responses. This strategy is motivated by the growing global

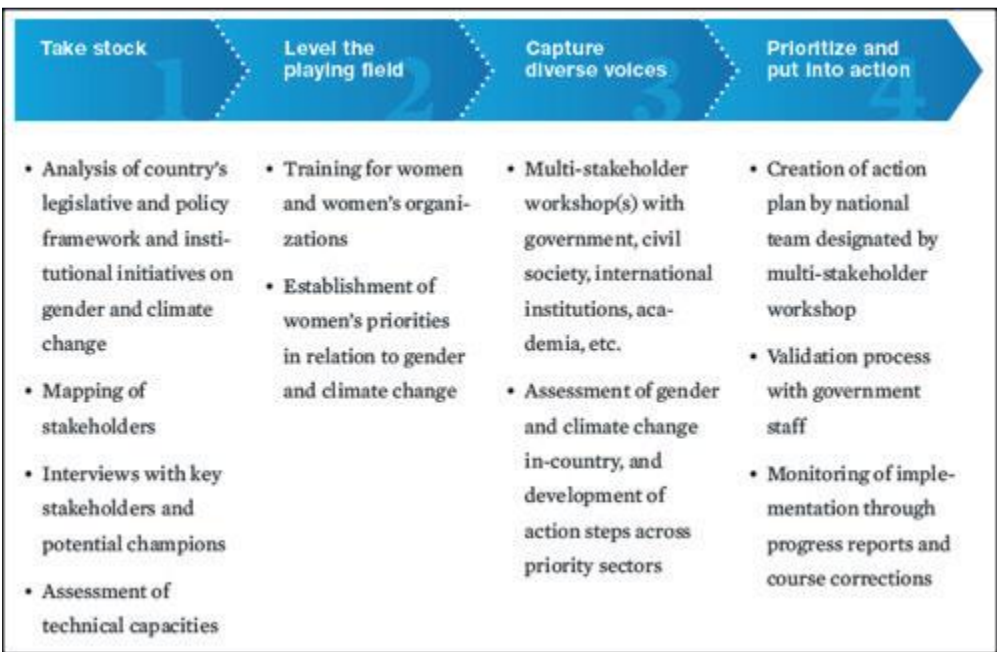
recognition of the importance of a gender dimension in climate change and environmental decision making. CCGAPs were first developed in a couple of countries (in Nepal, Bangladesh, Liberia, Tanzania, Jordan, Egypt, Panama, and Haiti) beginning from 2010. In the process, IUCN engaged the support of the Women's Environment and Development Organization (WEDO) and the Secretariat of the Convention on Biological Diversity (CBD) to develop CCGAP. The Global Gender Office of IUCN lead a process for revisiting and enhancing the Action Plan of the Gender, Environment and Climate Change Strategy developed and approved, as mentioned, in 2010.

Some of the specific methodologies being used in the GCCAP process include:

- **Participatory Needs Assessment:** A participatory needs assessment was conducted to identify the specific needs and priorities of women and men in relation to climate change. This was done through a series of consultations and workshops with a wide range of stakeholders, including government officials, civil society organizations, and community members.
- **Gender Mainstreaming:** The CCGAP process is designed to mainstream gender into all aspects of climate change policy and action. This means that the needs and priorities of both women and men are taken into account in all decision-making and planning processes.
- **Capacity Building:** The CCGAP process includes a number of capacity building activities for stakeholders. These activities are designed to help stakeholders understand the gender dimensions of climate change and to develop skills and knowledge to implement the GCCAP.
- **Monitoring and Evaluation:** The GCCAP process includes a monitoring and evaluation system to track progress and identify areas for improvement. This system will help to ensure that the GCCAP is implemented effectively and that its goals are met.

The CCGAP is a comprehensive and ambitious plan that has the potential to make a significant difference in the lives of women and men in Mali. By mainstreaming gender into climate change policy and action, the government of Mali is taking steps to build a more resilient and sustainable future for all. This Mali NCCGAP would be drafted on the basis of an analysis of the current national priorities concerning climate change and agriculture and by means of multi-stakeholder workshops. The detailed activities are shown in Figure 1 below.

FIGURE 1: STEPS TOWARD THE DEVELOPMENT OF A CCGAP



The plan aims to mainstream gender in all aspects of Mali's climate change policy and actions and is based on the principles that gender equality is a human right. It recognizes that women and men have different roles and responsibilities in the society and are affected differently by climate change as they both have different capacities to adapt to climate change.

The Mali ACCGAP implementation is by the efforts of Mali government (Ministry of Agriculture, Ministry of Gender, Child and Social Development) and the support from the Food and Agriculture Organization (FAO). The ACCGAP was founded on stout rudimentary pillars of human right and social justice within the principles of social equity, equality, non-discrimination, sustainability, parity and inclusiveness of men and women actors in the agricultural system. The plan has robust developmental vision of 'a climate resilient agricultural sector that contributes to food security and sustainable development in Mali'.

1.3. Key objectives of the climate change and gender action plan (CCGAP)

The ACCGAP would pursue four key objectives specifically designed to promote gender equality in climate change. They include to:

- (i) Increase women and youths' access to land, water, and credit
- (ii) Improve women's knowledge and skills in climate-smart agriculture
- (iii) Strengthen women's participation in decision-making and policy-making related to climate change
- (iv) Promote the use of gender-sensitive indicators to monitor and evaluate the plan's implementation

1.4. Initiatives taken to mainstream gender in climate change policies and actions in Mali

Meaning: Gender mainstreaming in climate change is the process of ensuring that the needs and priorities of both women and men are taken into account in all aspects of climate change policy and action (See UN Women: <https://asianpacific.library/publicationsof-mainstreaming-gender-into-climate-change-and-disaster-risk-reduction-policies>). Gender mainstreaming concept addresses the different ways in which women and men are affected by climate change, as well as the different ways in which they can contribute to climate change mitigation and adaptation. Mali has taken initiatives for mainstreaming gender into climate change policies. This was justified by the fact that by addressing gender inequalities in access to land, water and technology, women get empowered to adapt to climate change impacts more effectively. There is increased resilience to impacts of climate change by communities when women, and men are able to participate equally in decision-making and access to resources. In this way, women are able to participate in climate-resilient economic activities, improve their livelihoods, and health of their families and, reduce poverty. Gender mainstreaming has been the primary methodology used to integrate a gender approach into development and/or environmental efforts, then it could easily be understood why it has been gaining in momentum among policy makers, international organizations and donors to intensify sustainable development implementation. Mainstreaming gender does not imply merely a one-off solution to issues of equity and equality by adding women's participation to existing strategies and programs. Rather, it seeks to transform unequal societies and institutional structures to realize the full creative and productive potential of women to reduce vulnerability and enhance efficiency and effectiveness of development projects and programs.

There are a number of initiatives underway in Mali to mainstream gender in climate change. These initiatives are helping to ensure that women and men are equally represented in decision-making on climate change adaptation and mitigation, and that their needs and priorities are taken into account in climate change policy and action. These include:

- (i) The development of a National Gender Action Plan for Climate Change, launched in 2017. The plan aims to ensure that gender is mainstreamed in all aspects of Mali's climate change policy and action.
- (ii) The creation of a Gender Unit within the Ministry of Environment and Sustainable Development. The unit is responsible for promoting gender mainstreaming in climate change policy and action.

(iii) The implementation of a number of gender-responsive climate change projects, such as strengthening the resilience of Women Producer Groups and Vulnerable Communities project, which is funded by the United Nations Development Program (UNDP).

The steps taken by Mali government to implement gender policies and actions in the climate change include:

- The Ministry of Environment and Sustainable Development which has conducted a number of trainings on gender and climate change for government officials, civil society organizations and community members.
- The Ministry of Agriculture (MOA) which has developed a number of projects to support women farmers in adapting to climate change.
- The Direction of Gender and development which aims to integrate gender in development initiatives.

SECTION 2: THE CASE FOR GENDER EQUALITY

Typically, women have globally less economic opportunities to improve their lives. Jacobsen (2002) points out the importance for development of addressing the specific disadvantages that men face. Mikkola, (2005) argues that much like in women's case, there are global problem areas for men, in both developing and developed countries. The author reports data on the destruction of human capital that affects men more than women and points out how the changing roles of men put new demands on them. "Even if men's restrictions are perhaps more of a self-imposed nature. They should be paid attention to and targeted by policy measures" rightly thinks. She further stressed that there may be little hope of ultimately addressing women's issues if the men are ignored. The study suggests that literature review on gender equality mostly focusing on women, should keep in mind the issues where gender biases show destructively in men's lives.

On the role of gender equality in the process of development and growth, Mikkola (2005) reported some facts and survey that found that women play changing roles in processes of a global change. Such changes may involve in innovative processes that respond to climate change menace which has worsened the vulnerability of engendered men and women groups. The emerging institutional and structural changes in agricultural system may as well explain the persistent lower agricultural productivity and unwanted changes in their roles within the agricultural system. Mikkola's literature (2005) further links economic development with inequality issues that relate values bothering on market access, gender specific market failures in finance, political decision making, cultural restrictions and roles, legal and inheritance laws, resource allocation and practices etc. The argument is that economic welfare and women's welfare are associated. Amartya Sen (1999). thinks that GDP in itself is not the ultimate goal but rather the

freedoms associated with it- freedom for men and women to make choices that influence their lives or their economic wellbeing. there are typically some restrictions on freedoms that are directly counterproductive for the future development.

Important Gender Framework - Hypothesis on gender equality and growth

Mikkola adopted a simple relevant framework considered suitable in describing Mali's real situation. The framework shows a hierarchical view of gender where men are above the women - women equally considering themselves inferior and less valued than men. It involves a hierarchy of power. Gender equality on the other hand, entails equal value and opportunities for both men and women. Our current study therefore traces how the gender hierarchy manifests itself in Mali today. Care Report in Mali (May 2013) on a study carried out among men and women (aged 18-59) finds out that 41% of female respondents claimed to have experienced physical violence while 34.9% of their male counterparts claimed having used physical violence against a female partner. Furthermore, 41% of the men and 24% women work in agriculture and farming, including casual work (petty trading and similar activities). The fourth Assessment IPCC Report supposes that gender would be among one of the chief socioeconomic factors that determines vulnerability to climate change impacts (UNDP. 2005, Human Development Report. UNDP. New York, USA).

At the global level, women now represent 40% of the global labor force, and more than half the world's university students. Women play a vital role in agriculture accounting for 43% of the global agricultural labor force. In Developing countries, women make up even more of the agricultural workforce with estimates ranging from 60 to 80%. Women in agriculture engage in all aspects of production from planting and harvesting to processing and marketing. They also play a key role in managing natural resources such as water and soil. The World Development Report 2012 stated that gender equality is a core development objective in its own right. Economist found that the growth in employment of women in developed economies in the recent years contributed more to global economic growth than did in China. In the U.S.A, a study by McKinsey established that women went from holding 37% of all jobs to nearly 48% over the past 40 years, with an associated productivity gains in women 's share of the labor market accounting for approximately 25% of U.S. GDP. That equals more than \$3.5 trillion – more than the GDP of Germany and more than half the GDPs of China and Japan. Empowering women to exercise leadership within their communities contributes to climate resilience, ranging from disaster preparedness in Bangladesh, Indonesia and Nicaragua, to better forest governance in India and Nepal, to coping with drought in the Horn of Africa.

SECTION 3: THE CLIMATE CHANGE ACTION PLAN DEVELOPMENT METHODOLOGY

The methodology used in Mali's Gender and Climate Change Action Plan (GCCAP) process is a participatory approach that involves a wide range of stakeholders, including government officials, civil society organizations, and community members. The process begins with a needs assessment to identify the specific needs and priorities of women and men in relation to climate change. The needs assessment commenced with extensive review of literature including three key national documents, Mali's Nationally Determined Contributions (NDCs), Agricultural Development Policy (PDA) and National Adaptation Program Action (NAPA), '*Programme d'Action National d'Adaptation*'. These documents identify the five priority areas of Mali agriculture which include: (i) improved water management (ii) improved crop varieties and livestock management. (iii) improved agricultural process (iv) strengthening early warning systems and, (v) increased capacity building. They also provide information on the policies, strategies, and programs and expectations of the Paris agreement to gender, agriculture and climate change adaptation and mitigation in Mali.

A 2-day workshop was conducted in Bamako, as a form of participatory strategy involving the active participation of 30 persons (24 men and 6 women) from selected stakeholders. The workshop involved a presentation entitled, "The Importance of Gender Mainstreaming into Agriculture and Climate Change Initiatives of Mali". The objective of this workshop was to identify and understand the linkages between gender and climate change, especially for more cohesive and strategic response. A good understanding of the concept of gender mainstreaming is the first and basic for all actors in the systems that fight climate change. The work presented overview of climate change and gender inequality in Mali agricultural sector and also gave a detail of the underlying benefits in mainstreaming gender into climate change adaptation and mitigation processes. The purpose was to ensure that the assessment is relevant to the needs of the target population and that the action plan which is elaborated using a participatory approach is feasible.

In order to identify the respondents, the research team contacted the Ministry of Environment, Sanitation and Sustainable Development who is the government agency responsible for climate change in Mali; National Agricultural Research System (NARS) in Mali and, the Directorate of Planning and Prospective of the Ministry of Agriculture who identified the appropriate respondent institutions to interview. The list of the important organizations were compiled from the information provided by these three institutions and consultations from key stakeholders. The respondent samples include the institutions working with AICCRA Mali in Agriculture and Climate Change. See the list of respondent organizations and participants attached in **Appendix 1**. A total sample of **45** respondents were interviewed by means of well-structured two-levels questionnaires of general questionnaire and interview guide. The

general questionnaire was in form of a digital kobotool and a manuscript type. To ensure quality data collection process, there was training of 7 enumerators (2 women and 5 men) who conducted the data collection exercise. Extra effort was made to ensure fair representation of men and women respondents. This is to enable extraction of more features that will give better understanding of the problem we seek to solve as well as yield more robust model and more accurate predictions. Data collection was done in two levels. First, personal interview granted to the Heads of institutions, Director of Research, Gender focal points, agronomists, policy experts, etc. The second level of interview was to collect data from individuals and groups using key informant interviews, focus groups, and surveys.

The study seeks to provide answers to the following general questions: What are the current capacities of institution and organizations to address climate change and gender issues in the agricultural sector? What are the specific capacity gaps that need to be addressed in order to effectively implement climate-smart agriculture and gender responsive agricultural development? What are the most effective ways to deliver capacity building to the identified stakeholders in the agriculture sector? What are the challenges and opportunities for integrating gender into climate change adaptation and mitigation efforts in the agriculture sector? How can we ensure that capacity building initiatives are inclusive and accessible to all stakeholders including women and marginalized groups? What are the indicators to monitor progress in mainstreaming gender into climate change initiatives. Based on the findings of the analysis, the assessment team would develop recommendations on how to address the needs of the target beneficiaries. The findings would be disseminated to the stakeholders, so that they can be used to inform decision-making and program planning.

The information gathered from the data analysis would be summarized into a capacity check-list with a plan of action for capacity building. To ensure that this list is encompassing, it would cover the five objectives outlined in the Agriculture, Climate Change and Gender Action Plan (ACCGAP). The aspects of capacity building for stakeholders would be discussed in Sections 2 to 5 as follows: Section 2, Gender Responsive policy and planning: This involves ensuring that climate change policies and plans are gender-sensitive and that women's voices are heard in decision-making. Section 3: Climate change literacy: This involves providing stakeholders with the knowledge and understanding of climate change, its causes and impacts. Section 4: Gender Analysis: The gender analysis involves understanding the different ways climate change affects women and men, and how these impacts are shaped by gender norms and relations. Section 5: Adaptation and mitigation Strategies: This involves developing and implementing strategies to adapt to the impacts of climate change and mitigate greenhouse gas emission.

The capacity needs assessment would be followed by a series of consultations and workshops to develop a plan of action that addresses these needs and priorities. The plan would be implemented through a number of projects and activities that are designed to empower women and men to adapt to climate change and build a more sustainable future. The participatory approach being used in the CCGAP process ensures that the plan is relevant and effective. A

wide range of stakeholders would be involved to enable the government to gather valuable input and feedback that can be used to shape the plan. This approach also helps to build ownership of the plan among stakeholders, which is essential for its successful implementation.

Work In progress

A second multi-stakeholder workshop will be held in Mali to present the gender and climate change action plan, solicit feedback from stakeholders, and amend the gender and climate change action plan in light of the feedback.

This workshop is essentially meant to complete the zero-draft of the NNCCGAP. We will need to conduct a validation process in selected regions of Mali in order to harvest key valuable experiences and lessons learned in the field of agriculture, gender and climate change, from the various agricultural related projects and programs across Mali. The outcome of the participatory multi-stakeholder process would yield the much-needed draft Action Plan. The plan would show the roles that the various ministries and bodies play in initiating and facilitating efforts internally, and/or those of the strategic partners at the national, regional and international levels, to enhance gender-responsive climate change action. The due process being followed is outlined in the National Strategy of Adaptation and Mitigation of Climate Change, in complement to the NDC, NAPA and other sector-specific policies. Importantly, it also seeks to take advantage of opportunities that promote gender equality and facilitate transformational change in building climate resilient communities.

SECTION 4: COUNTRY OVERVIEW: THE CONTEXT FOR ADDRESSING GENDER AND CLIMATE CHANGE IN MALI

Mali is a west African landlocked Nation and a developing country. The country can be divided along a north-south axis., with the northern areas extending deep into Sahara and Sahel, and the southern region, which experiences a wetter, more tropical climate and where most of the country's economic activity is concentrated^{12,13}. Majority of the inhabitants live in the southern part of the nation which features Niger and Senegal River.

Mali is home to the inner delta of the Niger River. The country is bordered to the North by Algeria, to the East Niger, to the south Burkina-Faso and Mauritania and to the Southwest Guinea. There are four climate types distinguished in Mali. The population of Mali is about 19million on the land mass of 1,241,238 KM2 with only 14% of the land area suitable for agriculture this population is projected to 44 020,000 by 2050 (UNDESA (2015))². The Majority of the population lives in the rural area with only about 15% living in the urban area. The delta region of Mali is home to about 40% and the rural area 59% of the country's population. The floodplain region is under a lot of pressure from the high

population density, which can lead to problems such as environmental degradation, conflict, poverty etc. Although the government of Mali is working to address these problems but it may be more challenging if gender responsive policies are not applied in order to reach the more vulnerable, voiceless sub-population.

The high dependency on agriculture and concentration along floodplain and rural areas are among the factors positioning Mali as one of the most vulnerable countries to climate change in Sub-Saharan Africa and the wider world. The country is already experiencing effect of climate change including more frequent and severe droughts, floods and heat waves. Mali's physical vulnerability is heightened by socioeconomic and environmental factors, mainly: 1) the dependence on rain fed agriculture; 2) a high poverty rate and low Human Development Index; 3) high annual population growth rate (over 2.5%) that leads to increasing number of more vulnerable people settling in flood plains, combined with weak land use planning; 4) environmental degradation^{14,15} and 5) weak governance.

According to the 2020 Notre-Dame Global Adaptation Initiative (ND-GAIN) climate change vulnerability index, Mali is ranked 10th out of 181 countries in terms of climate change vulnerability. This rates Mali as one of the most vulnerable countries in the world to impacts of climate change based on exposure to climate stressors (such as drought, floods and heat waves); sensitivity or people's coping ability to climate change impacts and adaptive capacity of people and institutions to the impacts of climate change.

The climate change stressors are already leading to failure in livestock and crop production levels leading to food insecurity, disease, floods and conflicts (due to scarce water, land and other productive resources) displacing people and disrupting economic activities. Women and girls remain the most hit in all this crisis. The effect of climate change is expected to increase in future.

Nevertheless, Mali is not keeping quiet about all this. The government with support of various national and international institutions have embarked on proactive developmental and growth processes especially in improving agriculture, developing renewable energy sources, building infrastructures, promoting education and share of women in policy, governance and conflict management processes. Mali is not relenting in her efforts towards meeting the Millennium Development Goals, it has made and is determined to integrate major policies and programs, especially on climate change and gender mainstreaming for enhancing adaptive capability, development, and improvement of wellbeing, Mali is a carbon sink.

- **Topography:** Mali is a Sahelian, landlocked country, with almost two thirds of the total land area being covered by desert and semi-desert areas, unsuitable for agriculture. Mali's topography is somewhat a valuable resource for the country. It provides land for agriculture, water for irrigation and minerals for export. The topography is characterized with plateau and plains which are crossed by two of Africa's major river

systems, the Niger and the Senegal constituting significantly to the ground and surface water resources. The Inner Niger Delta, extends over more than 30,000 km², and is a special ecological site with significant agro-sylvo-pastoral potential. The surface water is mainly made up of the Niger and Senegal Rivers and their tributaries, offering an irrigable potential estimated at more than 2.2 million ha.

The southern part of Mali is characterized by a savanna landscape. The Niger River flows through this region creating a fertile flood plain. The central part is the Sahel, a semi-arid region with sparse vegetation. The northern part is the Sahara desert. At the northeast is the highest point of sand stone plateau, Hombori Tondo, which rises to an elevation of 1,155 meters. Mali's topography has a significant impact on the country's climate. The southern Savanah region is hot and humid during rainy season (June to May), the central Sahel region is hot during the dry season (October to May) and the northern Sahara Desert is extremely hot and dry year-round. The fertile flood plain of the Niger River is the most agricultural region in Mali. The central Sahel region is used for grazing but the north Sahara Desert is not suitable for agriculture or livestock grazing but it is rich in mineral resources such as gold, uranium and salt (Source: www.britannica.com/p). About 43.7 million ha of land can be used for agriculture and livestock. Only 5.2 million ha (11.9%) of the arable land are cultivated annually, of which less than 300,000 ha benefit from sources of water other than rain, and 120,000 ha benefit from irrigation water control. It has been estimated that due to climatic changes over the region, the major river basins could diminish significantly, with particular severity during the dry season and implications for fisheries.

- **Land use** - Agricultural production systems: There are four agro-ecological zones (AEZs) which can be identified in Mali: (i) The Saharan which is the largest zone covering 51% of the country's total area. It is covered by desert. The main agricultural activities are nomadic stock raising and few households cultivate sorghum around ponds. (ii) The Sahelian zone: This is divided into Sahelian-Saharan area in the North and Sahelian-Soudanian area in the South. It has average annual precipitation of 200 mm; covers 285,000 km² and occupies 23% of the country's land area. The main agricultural production involves cereal production (rice and sorghum), nomadic and transhumant livestock (camels, cattle, goats, and sheep). (iii) The Soudanian zone: this covers 17% of the country's land area and receives annual rainfall ranging from 600 and 800 mm. The agricultural crops produced are cereals (millet, sorghum, maize, and cowpea), groundnut and sesame as major cash crops. The smallholder farmers mainly practice sedentary livestock rearing and the area is highly valued for transhumant herding. (iv) The Soudan-Guinean zone covers only 75,000 km² of the country's land area. The rainy season lasts for 5-6 months with average annual precipitation varying between 800 and 1300 mm. Livelihoods in this region are based on maize, sorghum, cowpea, fruits and groundnut. Groundnut and maize are produced for household consumption and cash generation. Tubers and vegetables

(cabbages, okra, tomato, onion) are also grown for income diversification, The majority of farm households in this zone also practice sedentary stock rising for mainly cattle, sheep and goat. During the dry season, fields and pastures are exploited by transhumant herders from northern regions. Physically, Mali' agricultural sector's potential and assets are enormous. Pasture areas are estimated at more than 30 million ha, of which only nearly 1/3 is exploited, mainly because of insufficient access to water and various losses, particularly through trampling, fires and the effects of rangeland erosion. The herd is one of the largest in the sub-region and, is made up of mainly cattle, sheep-goats, camels and traditional poultry. Land not occupied by crops is generally dedicated to sylvo-pastoral activities, representing around 49 million ha. Livestock is mainly based on the exploitation of farm residues and natural resources. Poor management of land and natural resources, combined with harsh climate conditions (drought) have caused decreases in grazing land area, further inducing migration and even livestock losses.

- **Development;** According to the UNDP's Human Development Report 2021, Mali suffers from very low Human Development Index (HDI) of 0.428, ranking 187 out of 189 countries. The country's HDI is challenged by economic and non-economic factors, such as poverty, inequality, conflict, and climate change. According to the World Bank, 44% of Malian households live below the national poverty line. Inequality challenge in Mali is evidence in 10% of Malians controlling more than 40% of the country's wealth while larger proportion are struggling to make ends meet.

However, it is worth nothing that, despite these challenges, Mali has made significant progress in human development in recent decades. The country's HDI has increased by more than 20% since 1990, it is yet lower than the HDI of 0.692 for developing countries and the 0.55 HDI average for SSA in 2021. The life expectancy has increased from 48 years to 59 years over the same period due to factors like increased investment in education and healthcare, economic growth and political stability. These growth and development gains contribute to progress in achieving targets identified in the Millennium Development Goals (MDGs).

The government of Mali has a number of policies and programs in place to address the challenges to human development. They include the national poverty reduction strategy, which aims to reduce poverty by 50% by 2030; The National Gender Policy, which aims to promote a gender equality and women's empowerment; the National Peacebuilding Strategy, which aims to build peace and stability in Mali; the National Climate Change Adaptation Strategy, which aims to adapt to the impacts of climate change.

- **Economy;** Mali is a landlocked country with population of over 20million people. The country is a low- income country with a GDP per capita of USD 1,200. Mali's economy has been struggling since the 2012 civil war

when its GDP growth rate was -1.2%, the lowest in over a decade. The economy is improving gradually but the growth remains below pre-conflict level. The economy is based on agriculture, livestock and mining. Agriculture is the largest sector of the economy accounting for over 40% of GDP and employing over 80% of the workforce. In 2022, GDP growth is estimated at 1.8%. Mali is a major exporter of cotton, as well as a significant producer of rice and livestock. Mining is the second largest sector of the economy, accounting for over 15% of GDP. Mali produces bauxite, copper iron ore and major producer of gold. Service sector accounts for 45% of GDP (Mainly trade, transportation, and tourism). Mali's economy has been growing in recent years with average GDP growth rate of 5% per year. Mali is currently one of the poorest countries in the world with poverty rate of over 40%.

Although Mali has observed a significant growth rate in its GDP, the economy has suffered because of insecurity, climate change and high level of poverty among others. Mali has experienced significant decline in rainfall recently and this trend is expected to continue. Climate change is affecting Mali's agricultural sector in diverse ways. (i) Decreasing agricultural productivity through decreased rainfall (ii) causing crop failures and food shortage through severe drought (iii) leading to loss of agricultural land and the displacement of people through desertification (iv) Damaging crops and infrastructures and impaired health through extreme weather events such as floods and heat waves. This has led to increased droughts desertification and crop failures. (v) Mali is facing security crises in the north and center of the country, which has disrupted economic activities and has led to a decline in foreign investment. The Malian government is taking steps to address the challenges posed by climate change. The government has developed a national climate change strategy, which includes measures to improve agricultural productivity, adapt to climate change, and mitigate greenhouse gas emissions. Despite these efforts, Mali remains highly vulnerable to the impacts of climate change. The country's economy and food security are at risk, and the government will need to invest in climate adaptation and mitigation measures in order to protect its population.

World factbook has shown that climate change is a threat to Mali's economy and food security. The government is taking steps to address this challenge, but more needs to be done. The international community also needs to provide supports to Mali as it adapts to climate change.

SECTION 5. GENDER INEQUALITY

Gender inequality is a major challenge in Mali's agricultural sector. Women make up the majority of agricultural workers in Mali but they face significant barrier to access production resource making women farmers often less productive than male farmers. The 2020 Gender Inequality Index (GII) published by the UNDP assigns Mali a value of

0.587, ranking it 184 out of 189 countries on the GII above some of its neighbors though slightly below the Sub-Saharan African average of 0.589. The GII for Mali's agricultural sector is 0.612. Though this is moderate, yet it is higher than the national average implying a significant gap between women and men in the sector in terms of opportunities and outcomes. This is because women farmers face additional barriers in the agricultural sector such as: Limited access to land due mainly to customary land tenure systems that favor men; and the fact that women are often excluded from decision making about land use; Limited access to credit which makes it difficult for women to invest on their own farms. In Mali, women are often seen as less creditworthy than men are and often lack collateral required to obtain loan; limited access to extension services which provides farmers with information and advice on agricultural practices. This is because extension services in Mali are often located in remote areas and they may not be tailored to the needs of women farmers.

Addressing gender inequality in Mali's agricultural sector is essential for the development of the country. By empowering women farmers and providing them with the resources they need, Mali can increase agricultural productivity, reduce poverty and improve food security which are at the heart of the country's economic development. The following actions need to be taken:

Women Situation

Women in Mali face significant challenges in all areas of life, including politics, society, economics and culture.

Politics: Women are underrepresented in Malian politics. In the 2020 parliamentary elections, women won only 27.3% of seats which is a slight increase from the 24.7% won in 2013. However, Mali still lags behind many other African countries in terms of women's political representation. For example, in Rwanda, women hold 64% of seats in parliament.

Society: Women in Mali are often seen as inferior to men. They are expected to take on the majority of the unpaid domestic work and childcare and they have less access to education and employment opportunities. Women are more likely to be victims of violence, including domestic violence, sexual assaults, and female genital mutilations. In 2018 survey, 18.4% of Malian women claimed to have been physically or sexually abused by a partner.

Economics: Women in Mali are more likely to live in poverty than men. They also earn less money than men, and have less access to credit and other financial services. The 2018 Mali Demographic survey, provided that 44% of the female-headed households live below the national poverty line of 40% compared to 33% of the male's. In rural areas,

women are often the primary farmers. However, they often lack access to land, water and other resources. This makes it difficult for them to produce enough food to support their families.

Culture: Malian culture is patriarchal where men are seen as the head of the household and the decision-makers. Women are expected to be subservient to their husbands and fathers. This view of gender roles is often reinforced by religious leaders. Despite the challenges they face, Malian women are making progress in all areas of life. They are becoming more politically active, gaining access to education and employment opportunities. They are speaking out against violence and discrimination. There are a number of organizations working to promote gender equality and combat violence against women. Women's life expectancy is significantly higher than that of the men at birth in Mali at 60.3 years, compared with 57.6 years for the men. The country's maternal mortality rate is one of the highest rates in the world reaching 546 deaths per 100,000 live births in 2018. Literacy rates in Mali are low, especially for women. The 2017 Mali Demographic Survey reports that the male literacy rate reached 34.7% while females reached 26.7%. Girls average 3.2 and boys 4.2 years in total schooling.

SECTION 6: LEGAL FRAMEWORK: MANDATES AND POLICY RELATED TO AGRICULTURE

6.1. Mali's Nationally Determined Contributions (NDC)

The NDC is an embodiment of the efforts by UN countries to reduce national emissions and adapt to the impacts of climate change. Mali produced its NDCs to reduce greenhouse gas emission following the adoption of a new climate change agreement at 21st session of the COPs to the UNFCCC in Paris, in December 2015. Mali was among some 54 African countries who signed in NDC and the 48 countries (89%) ratified NDCs as of Oct 2021 (Africa NDC Hub). In the NDCs Mali outlines mechanism undertaken to fulfil objectives of the endorsed Paris Agreement. The Department of Climate Change of the Federal Ministry of Environment in a bid to achieve the NDC has developed the National Climate Change Strategy and Action Plans. Mali submitted to the UNFCCC its First NDC (slightly revised INDC) in September 2016⁵. It adopted NDC 2015-2019 that was later reviewed in 2019 to the current version (2020-2030). In its NDCs Mali presents itself as being highly vulnerable to the negative effects of climate change, resulting in over exploitation of natural resources and land degradation⁶. The first plan (2015-2019) included three sectors (energy, agriculture, and forestry) and this revised version, (2020-2030) incorporated Gender to it.

Mali's revised NDC (2020-2030) was based on outlining gender considerations and underlining the need to adopt a gender-sensitive approach to climate change policy-making and programming in Mali. It was designed to correct the

initial wrong assumption that the negative impacts of climate change had similar effects on women and men while adopting a generic approach in the mitigation. Now, the NDC recognizes that, 'women and men experience climate change differently'. Mali has demonstrated her unwavering strategic intent to mainstream gender in the objectives of adaptation through their NDCs and identified their adaptation priorities through National Adaptation Plan (NAP) processes. Through this process, Mali identifies addresses and reviews her adaptation needs, and advances implementation of adaptation measures. "Mali's revised NDC has focused on gender through which Mali proposes the implementation phase, to be a question of integrating gender indicators and adopting a gender-sensitive approach in the development of policies, programs and projects contributing to the achievement of the objectives of the NDC.

Mitigation. Mali is committed to reduce the GHG emissions by 27% by 2030, compared to a Business-as-Usual (BAU) scenario targeting 29% and. 31% reduction in agricultural and energy sectors respectively. Mali updated the baseline period with mitigation scenario GHG reduction levels relative to the baseline scenario of 31% for energy, 25 % for agriculture, 39% for forests and 31% for waste. The synthesis of emissions and sequestrations by 2030 gives a reduction of 40% of these total emissions. The conditional costs of mitigation measures are estimated at \$34.68 billion for 2015-2020. Based on the evaluation of the first NDC, with the climatic emergencies observed at the global level, Mali has entered into a collective approach to raise its ambitions.

Adaptation: The NDC also includes a section on adaptation, in which Mali prioritizes the development of a green and climate smart economy. Focus sectors for adaptation are forestry, climate smart agriculture, renewable energy, pastoral management and Integrated Water Resources Management (IWRM). Estimated cost of adaptation measures are USD 1.06 billion, covering a period from 2015 to 2020. NDC prediction for adaptation depicts an inclusive approach involving mobilizing local communities, vulnerable communities and women's groups to improve their resilience to climate change; A series of sectoral projects in the fields of agriculture, civil society and capacity building and; the development of research on adaptation to climate change. The realization of Mali's predictions is conditional on funding. Mali is already receiving climate funding in accordance with the obligations of the UNFCCC, and the Paris Climate Agreement from different sources, including funds under the convention; the support of Technical and Financial Partners, bilateral and multilateral development banks, private sector etc. Mali communicated their vision and objectives on adaptation through the NDCs, and has developed policies, programs and strategic actions in the process to develop National Adaptation Policies.

6.2. National Adaptation Program of Action

A multi-sectorial group, with representation from the government of Mali, NGOs, CSOs, and humanitarian agencies, was established to create a National Adaptation Program of Action (NAPA) to identify priority activities and respond to reduce the risk of impact from climate change. The NAPA was developed in 2007 and updated in 2013. It identified the 5 priority areas in agriculture: (i) Improved water management (ii) Diversification of crops and livestock (iii) Improved agricultural process (iv) Strengthening early warning systems (v) Increased capacity building. In addition, other areas that are important for agriculture include reducing deforestation, improving soil fertility and increasing access to markets. The NAPA includes a number of measures to promote gender equality and empower women including (i) increasing women's participation in decision-making (ii) Providing women with access to resources (iii) Promoting gender sensitive climate change education.

6.3. National Action Plan (NAP) in conflict and climate change

Mali's National Action Plan (NAP) on conflict and insecurity is a plan to address the challenges posed by conflict and insecurity to climate change adaptation. The NAP was developed in 2017 and updated in 2022. It identifies several priority areas for gender action: (i) Increasing women's participation in decision-making at all levels from local to national for climate change adaptation and in other leadership positions. (ii) Providing women with access to resources. (iii) Promoting gender-sensitive climate change education.

6.3.1. The National Action plan on Security

The Resolution was adopted by the United Nations Security Council on 31 October 2000 and is an innovative international norm, which calls for the full and equal participation of women in all initiatives concerning peace and security. Mali has experienced an unprecedented socio-security crisis that has affected the entire population. Women and children have been the most affected and have been impacted by all manners of violence. NAP document acts as a reference tool for the consideration of specific rights and needs of women and girls in all peace, mediation and national reconciliation processes.

6.3.2. Mali Constitution

Gender is a priority recognized in the Preamble of the Constitution of the third Republic of Mali: “The sovereign people of Mali declared its determination to defend the rights of Women and Children as well as the cultural and linguistic diversity of the national community.”

The Constitution of Mali adopted in 1992, and amended in 1998 and 2000, is founded on the principles of equality of all citizens before the law regardless of gender, race, social status or origin.. It includes an Article of Equality (Article 2), which states, “Men and women have the right to participate in the political, economic, social and cultural life. Of the country”. Article 25 states that the State will take all necessary measures to promote gender equality and to eliminate all forms of discrimination against women.

6.3.3. National Communications to the UNFCCC

At the time of this CCGAP, Mali has submitted its first and second national communications to the UNFCCC and is currently working on the third.

SECTION 7. LEGAL CLIMATE FRAMEWORKS, MANDATES AND POLICIES FOR ADVANCING GENDER EQUALITY

7.1. Institutional arrangements

The government of Mali has a number of agencies that are responsible for agriculture and climate change including the Ministry of Agriculture, the Ministry of environment, the Ministry of Livestock and Fisheries and the Ministry of Water Resources, the National Agency for Sustainable Development (AEDD); the Climate Change Fund (FCF) for mobilizing financial resources to support adaptation and mitigation. The ministry has a number of specialized agencies such as the National Cereals Office (ONIC) and the National Livestock Office (ONAGREF). The Ministry of Agriculture, Livestock and fisheries (Ministere de l’Agriculture, de l’Elevage et de la Peche, MAEFR) is the government ministry responsible for Agricultural development and food security in Mali. The MAEFP has a number of departments that are specifically focused on gender and climate change. Including:

(i) The Department of Gender and Development (Direction de la Genre et de Developpment, DGD) is responsible for promoting Gender equality and women’s empowerment in the agricultural sector. (ii) The Department of Climate Change (Direction du Changement Climatique, DCC) is responsible for developing and implementing Mali’s national

climate adaptation and Mitigation strategy. The DCC is working to mainstream climate change adaptation and mitigation into agricultural planning and programming. The MAEFR works closely with other government ministries, civil society organizations, and development partners to address the challenges of gender in the agricultural sector being the the lead of all the agencies responsible for developing and implementing agricultural policies, providing technical assistance to farmers and regulating the agricultural sector.

A number of government organizations (NGOs) and non-governmental organizations of the civil society in groups (CBOs) who play a significant role in Mali's agricultural sector. NGOs provide a variety of services to farmers including agricultural extension, input supply and market access. CBOs also play substantial role in mobilizing farmers, providing sustainable agricultural practices, and managing natural resources. Research institutions may be governmental or non-governmental. Among the legal frameworks and policies developed by the Malian government are the National Climate Change Action Plan (2023-2025), the National Agriculture Sector Development plan (2021-2020), the National Forestry Action Plan (2013-2025).

7.2. Global climate policy

Mali signed the United Nations Convention on Climate Change (UNCCC) on December 28, 1994 and ratified it on May 19, 1997. At the first Conference of Parties, in Berlin in 1995, a Special Group was established for negotiating a Protocol to the Convention. The protocol was adopted on December 1997 in Kyoto and was named the Kyoto Protocol. Mali signed the Kyoto Protocol on January 27, 1999 and ratified it on March 2002. The Protocol entered into force on February 16, 2005.

Mali falls into the group of developing countries whose objectives were not reducing their glasshouse gas emissions rather they faced the objective three of the Kyoto protocol otherwise referred to as the objective of the "Clean Development Mechanism". This objective is to assist this group of Parties in achieving a sustainable development as well as in contributing to the ultimate objective of the convention, and in assisting in helping, the parties achieve of the convention, and in fulfilling their figured commitments to limit and reduce their glasshouse gas emissions. The Kyoto Protocol urged 'Parties to take the measures necessary to enable women to participate fully in all levels of decision making relevant to climate change' and invited Parties to consider the nomination of women for elective posts in any body established under the Convention or the Kyoto Protocol. The following is a summary of the key decisions, highlighting the chronology of gender and climate change at the UNFCCC:

Table 1: Key global gender and climate change decisions

Year	Cop	Decision
2001	Cop -7	Gender in the NAPs: Decision 36: “Improving the participation of women in the representation of Parties in bodies established under the UNFCCC and the Kyoto Protocol”.
2007	COP-13,	Held in Bali, Indonesia and, clearly underlined the commitments of Member States concerning climate change. Parties also formulated the Bali Action Plan, a crucial framework to begin negotiations to reach a global agreement on climate change by the end of 2009. The Bali Action Plan affirmed that an effective approach to climate change includes mitigation, adaptation, technology and financing. The Bali conference led toward the promotion of gender equality in the framework of the UNFCCC.
2010	Cop- 16	Gender in the Cancun Agreement/ Shared Vision: The Preamble of Decision 1/CP.16 notes resolution 10/4 of the UN Human Rights Council on human rights and climate change and notes differential impacts of climate change on segments of the population, owing to intersections such as age and gender. Additionally, in the operative section on ‘Shared Vision’, the decision recognizes that “gender equality and the effective participation of women, are important for effective climate action on all aspects of climate change”. The key decisions are: -Improving gender balance & increasing women’s participation in UNFCCC processes, including bodies constituted under Convention, Kyoto Protocol & Paris Agreement -Increasing awareness & support for development & effective implementation of gender-responsive climate policy and action at regional, national & local levels
2011	Cop- 17	Green Climate Fund (GCF): The GCF governing instrument states that the fund will take a gender-sensitive approach. The GCF board will give due to consideration to gender balance and develop mechanisms to promote the input and participation of stakeholders, including private-sector actors, civil society, organizations, and vulnerable groups, including women and indigenous people. The GCF secretariat will also take into account gender balance.
2012	Cop- 18	Decision 23 Adopted in Doha: “Promoting gender balance and improving the participation of women in UNFCCC negotiations and in the representation of Parties in bodies established pursuant to the Convention or the Kyoto Protocol.”
2014	Cop-20	Decision 18: Lima Work Program on Gender: “... enhance the implementation of decisions 36/CP.7, 1/CP.16 and 23/CP.18 by inviting Parties to advance gender balance, promote gender sensitivity in developing and implementing climate policy, and achieve gender-responsive climate policy in all relevant activities under the Convention.
2015	Cop- 21	The Paris Agreement: Preamble: “Acknowledging that climate change is a common concern of humankind, Parties should, when taking action to address climate change, respect, promote and consider their respective obligations on human rights, the right to health, the rights of indigenous peoples, local communities, migrants, children, persons with disabilities and people in vulnerable situations and the right to development, as well as gender equality, empowerment of women and intergenerational equity.” Article 7: “Parties acknowledge that adaptation action should follow a country-driven, gender-responsive, participatory and fully transparent approach...” Article 11: “Capacity-building should be guided by lessons learned, including those from capacity-building activities under the Convention, and should be an effective, iterative process that is participatory, cross-cutting and gender-responsive.”
2016	Cop- 22	Decision 21: Extending the work of the Lima Work Program on Gender for three years, to be reviewed at COP25 in 2019, as well as the following new activities:

		• Requesting both technical bodies and Parties, as well as the Financial Mechanism, to enhance communications and reporting on progress implementing gender-responsive climate policy; • Requesting that a gender perspective be considered in the organization of the technical expert meetings (TEMs) on mitigation and adaptation; • Requesting Parties to integrate local and traditional knowledge in the formulation of climate policy and to recognize the value of the participation of grassroots women in gender-responsive climate action at all levels; • Inviting Parties to appoint and provide support for a national gender focal point for climate negotiations, implementation and monitoring. Mandated Parties to develop possible elements of the gender action plan for consideration at SBI47 (COP23)
2017	COP -23	Decision -/CP.23: The Gender Action Plan: “To support the implementation of gender-related decisions and mandates under the UNFCCC process, which may include priority areas, key activities and indicators, timelines for implementation, responsible and key actors and indicative resource requirements for each activity, and to further elaborate its review and monitoring processes;” “Gender-responsive climate policy requires further strengthening in all activities concerning adaptation, mitigation and related means of implementation (finance, technology development and transfer, and capacity-building) as well as decision-making on the implementation of climate policies. The GAP recognizes the need for women to be represented in all aspects of the UNFCCC process and the need for gender mainstreaming through all relevant targets and goals in activities under the Convention as an important contribution to increasing their effectiveness.”
2018	COP-23	Decision 3/CP.23: Parties adopted a multi-year Gender Action Plan (GAP) with an initial focus at COP24 on monitoring and reporting of gender-disaggregated climate change impacts (UNFCCC, 2018)
2019	3/CP.25	Enhanced LWPG and its Gender Action Plan (GAP)- The enhanced GAP sets objectives & activities under five priority areas: a) Capacity-building, knowledge management & communication; b) Gender balance, partnership & women’s leadership; c) Coherence; d) Gender-responsive implementation and means of implementation; and e) Monitoring & reporting.

CCAFS (2020) Working Paper No. 323, CGIAR Research Program on Climate Change, Agriculture and Food Security (CCAFS).

7.3. Regional climate policy

Mali is a member of African Union (AU) and the Economic Community of West African States (ECOWAS) both of which have adopted climate change. The 23rd African Union Summits of Head of States and Governments Decision on the High-Level Work Program on Climate Change Action in Africa (*WPCCAA*) *Assembly/AU/Dec.538 (XXIII), Malabo 2014* recognizes the need for the elaboration of gender and youth engagement within the climate change negotiations. This led to the Committee of African Heads of States on Climate Change (CAHOSCC), Gender, and Youth Meeting on Climate Change, organized by the African Union Commission and United Nations Economic Commission for Africa (UNECA) in Nairobi, Kenya, from June 7 – 10, 2016. Others include :-Convention on the Elimination of All Forms of

Discrimination Against Women (CEDAW); -Maputo Protocol on the Rights of Women in Africa Mali adopted the Detailed Program for the Development of Agriculture in Africa (CAADP) at the Maputo summit in July 2003 and signed, within the framework of its implementation, the agreement which commits the countries of the African Union to devote at least 10% of their national budgets to agricultural investments with a view to achieving an agricultural growth rate of at least 6% per year. The country has also subscribed to the principles of the Paris Declaration on Aid Effectiveness, set out in 2005 by Development Partners. -The African Union Climate Change and Resilient Development Strategy and Action Plan 2022-2032 provides framework for ensuring climate justice for Africa through inclusive & equitable participation in climate action & climate-resilient development pathways. Fifty-four (54) African countries signed; 48 countries (89%) ratified NDCs as of Oct 2021 (Africa NDC Hub).

7.4. National policy initiatives

The Malian government has recognized the need to address the gender dimensions of climate change. To this effect, the NCAGF was developed by the Government of Mali in collaboration with a number of development partners, including the UNDP, FAO and the World Bank. The NCAGF principle on gender is based essentially on gender equality for sustainable development. Some of the policies include:

1. The Mali's revised NDC (2020 -2030) included gender equality considerations. Mali referred to gender in climate finance strategies, policies, strategies and framework. Mali highlighted the importance of women and girls' climate change adaptation. This was not recognized in the first generation of the NDC plan.
2. The National Action plan on Security; The Resolution was adopted by the United Nations Security Council on 31 October 2000 and is an innovative international norm, which calls for the full and equal participation of women in all initiatives concerning peace and security. Mali has experienced an unprecedented socio-security crisis that has affected the entire population. Women and children have been the most affected and have been impacted by all manners of violence. NAP document acts as a reference tool for the consideration of specific rights and needs of women and girls in all peace, mediation and national reconciliation processes.
3. Mali's Action Plan for 2019-2023 is the Government of Mali's response to the United Nations Security Council resolutions on the Women, Peace and Security agenda. It includes emerging issues such as the impact of radicalization and violent extremism against women and girls, the situation concerning displaced women and girls, the role of men in advancing gender equality and women's resilience to climate change. NAP1325 will be implemented in collaboration with national and international partners. It acts under the direction of the MPFEF alongside representatives of fifteen ministries and other bodies like the National Assembly, government programs,

Civil Society Organizations (CSO), the United Nations System (UNS) and technical and financial partners (TFP). NAP is specifically aligned with the Agreement for Peace and Reconciliation in Mali resulting from the Algiers Process, the Government's national strategies and policies in relation to gender, gender-based violence, economic recovery and sustainable development among others.

Table 2: Other gender Relevant National policies and their objectives

	Policies\strategies	Objectives
i	National Gender Policy	Improving the involvement of women in the country's management process
ii	National Agriculture Policy (PDA)	Promote sustainable, modern and competitive agriculture based primarily on family farms
iii	National Planning Policy	Give economic development planning a territorial dimension, within the framework of an organization of the territory
iv	National policy of Decentralization space taking into account decentralization	Strengthen the process of democratization of society and adapt the missions and organization of the State to the requirements of promotion of local initiatives
v	National population policy	Mastering and managing the evolution of the population of Mali

SECTION 8. CHALLENGES, RESPONSIVE POLICIES AND IMPLEMENTING STRATEGIES AND PROGRAMS ON GENDER AND CLIMATE CHANGE IN AGRICULTURE

8.1. Challenges for mainstreaming gender in agricultural and rural development

There are four major gender challenges affecting the mainstreaming of gender in agricultural and rural development in Mali as follows.

8.1.1. Women's limited access to resources necessary for agricultural production

Women have limited access to necessary agricultural development resources such as natural resources, inputs, and extension services. In Mali and other African countries, over 50% of the rural farmers contribute to agricultural production, food security, and development of the rural economy (FAO, 2011). Discriminative social norms, laws, and regulations about women's actions and roles, limit their access to productive resources and opportunities and impair

their capability to contribute to rural development. Only 3.1% of these women are agricultural landholders in Mali. Land ownership is an invaluable productive factor as it is often the condition for membership in farmer organizations (example irrigation and water user associations, and producer associations). Most extension services and projects target only association members and as such non-members are ineligible to benefit from such services.

In African households, societal norms preconceive men as breadwinners and women as mere supplementary workers to men, which is the taproot of inequality. Strikingly, only male household heads own and manage household assets and agricultural resources, and take major resource management decisions (e.g. deciding on crop varieties, types of fertilizers and chemicals, or the timing of their application). In such cases, women cannot use their knowledge and experience for productive activities. Men are often associated with professional knowledge and skills while women face challenges in access to training and extension services to learn skills to use machines, apply improved management practices, and vaccinate animals. Most agricultural machines are designed to fit men and are not women friendly due to the greater physically effort required. Lack of training and extension services about agricultural technology and machines meeting women's needs is another factor that hampers women's participation. Women's tight schedule discourage them from trainings because training sessions are often conducted during the time when women are busy doing housework.

8.1.2. Unpaid care and domestic work limiting women in securing time for productive activities

Women assume responsibilities and roles that attract no economic remuneration - not paid for care and domestic work, such as household chores, child rearing, attending family needs, and caring for the sick - in addition to agricultural work and small businesses. Globally, women spend three times as much time as men do on unpaid care and domestic work (UN Women, 2019). The unpaid heavy workload of the rural women demand more time and energy, and time on community activities. Women undertake the roles of the men when they migrate.

8.1.3. Women's limited participation in decision-making

Despite rural women's important roles in agricultural production, their participation in decision-making tends to be limited at the household, community, and administration levels. Men are leaders of communities, producer groups and cooperatives, excluding women from decision-making. Farmer organizations tend to have a small number of female members; because most often only household heads may belong making voice of women not heard or

reflected in-group decisions. Men occupy most of the important positions of local and central government organizations, such as ministries and research institutions, because of the perception that men should take responsibilities of technical and professional work and that women lack the capacity to perform the duties due to missing opportunities to receive professional education. Policies, plans, and technologies tend to be developed on the vague assumption that farmers are men despite the fact that about the half of farmers are women. Officials and extension workers who disseminate agricultural technologies and policies to farmers also tend to be men. As a result, little attention has been paid to the situations, issues, and needs of women farmers (UN Women, 2019).

8.1.4. Sexual and gender-based violence

Rural areas face challenges of traditional customs and, sexual and gender-based violence (SGBV). Globally, Violence against women is a common issue across cultures and societies, and especially in rural areas where the norms is that men should control and women obey. This is where serious issues of SGBV and farmer clash occur. Girl child is given to marriage, domestic violence (DV), and sexual violence, occur which harm the physical and mental health of women and girls and hinder their social and economic participation. Girl child suffers challenges of early marriage, unwanted pregnancy and dropping out of school because of social norms. Many teenage girls drop out of school because of pregnancy and childbirth and are deprived of education. Violence against women and girls causes severe social and economic damage to rural villages. The cost caused by violence against women and girls is estimated to account for 2% of the GDP of a country (UNFPA, ICRW, 2009).

8.2. Strategies for mainstreaming gender in national climate change plans

8.2.1. Gender mainstreaming strategies in agriculture

A good understanding of the concept of gender mainstreaming is the first and basic for all actors in the systems that fight climate change. GM is one concept with diverse but convergent point of view. In terms of development projects, it refers to the incorporation of a perspective that promotes gender equality and women's empowerment in all stages of a project (i.e., planning, implementation, and monitoring and evaluation). Mali acknowledges that gender mainstreaming will contribute to the protection of human rights of the women and diverse people, and allow all people engaged in agriculture to achieve their full potential as producers or value chain actors. Hence, the country has received international and internal supports of funds and technical aids to facilitate agricultural and rural

development and realize concrete outcomes through GM and has made solid moves toward realizing its target including: Important steps taken:

(1) Mali's NDC adopted decentralization of power and inclusion of stakeholders in its strategies. This innovative approach takes into account regional specificities as part of a participatory and inclusive approach. Decentralization is about recognizing and involving local authorities in the decision process to tackle climate change. By this means, Mali adopted territorial approach through regional consultations seeking to integrate resilience into regional and local development. In effect, National policy of Decentralization space is taking into account - strengthening the process of democratization of society and adapting the missions and organization of the State to the requirements of promotion of local initiatives. Effective decentralization in the NDC provides for gender to be considered in all acts involving planning and budgeting in all developmental processes in the local areas. The consultation to develop the NDC was through an inclusive process that took place in five regions and the District of Bamako. This design involves civil organizations and other stakeholders in exchanges and discussions on the level of vulnerability of these regions in the face of climate change. They are also to consider the type of mitigation and adaptation strategy to put in place. This activity was funded by collaboration of regional actors, the technical and financial partners (UNDP) and, Environment and Sustainable Development Agency (AEDD).

2. One of the innovations of this NDC is the consideration of the gender dimension through all the sectors concerned. Hence, updating adaptation NDC was another gender mainstreaming strategy. The Mali Adaptation NDC of 2015 was updated in 2020 to inculcate gender considerations in policies and transverse gender across all sectors especially agricultural sector. The strategy was informed by recognizing that: (1) climate change has differential impacts on men and women; (2) men and women offer contributions different and complementary in the areas of mitigation and adaptation, (3) women and men should equally benefit from the opportunities offered by the control programs against climate change at the practical and strategic level (access to new technologies, new green jobs, etc.).

8.2.2. Development of adaptation plans and strategies

The summary of GHG emissions by sub-sector shows that the Agriculture sector represents 69% and the Livestock sector 31% of the total GHG emission in Mali. REDD+ potential is exploitable by the implementation of NDC 2020. One of the strategies is the involvement of the Non-state actors in 2020-2030 NDC. For this period, civil society (mainly NGOs) have been involved in planning several projects for the mitigation and adaptation actions. The adaptation programs and projects includes the Sub-regional Advocacy and Capacity Building Program, "Citizens' Way for Climate Change", Mali; Burkina, Support Program for Capacity Building for Adaptation to Climate Changes, Mali

(PARCAC) and Program to Strengthen the Resilience of Vulnerable Agro-Pastoral Communities in the Western Sahel to the Effects of Climate Change. All these plans are gender sensitive.

8.2.3. Responsive policies and implementing strategies and programs on gender and climate change in agriculture

To implement the objectives of the policies, Mali has designed a number of programs and initiatives. The National Gender Action Plan for Climate Change (NAP-CCC); the women's Climate Resilience Fund (WCFR); The Malian Women's Network for Climate Change (Reseau des Femmes Maliennes pour le Changement Climatique) The NAP-CCC is a National level plan that guides the implementation of the gender policy on climate change. It identifies the following priority areas for action: Gender mainstreaming in climate change policy and planning; women's participation in climate change decision-making; women's access to climate resilient resources and technologies; reducing women's vulnerability to climate-related disasters and promoting gender equality in climate change adaptation and mitigation efforts. The WCFR is a fund that provides financial support to projects and programs that promote women's resilience to climate change. The WCFR has supported a number of projects including a project to provide women with access to climate resilient agricultural technologies; a project to provide women with training on climate change adaptation and mitigation and a project to support women's leadership in climate change initiatives. The Malian Women's Network for climate change is a network of women's organizations that works to promote women's rights and empowerment in the context of climate change. The network has a number of activities that may include advocacy for gender-responsive climate change policies and programs; capacity building for women's organizations on climate change issues; networking and collaboration with other organizations working on climate change.

Some of the agricultural related strategies and plans which include GHG emission mitigation assessment are:

- The National Agricultural Sector Investment Plan (PNISA) 2015-2025: The Strategic Investment Framework for Sustainable Land Management; - the improvement of the mechanism for allocating iron and strengthening the strategic positioning of women in all links in the value chain of promising sectors in order to diversify sources of income and reduce pressure on agricultural land. The overall amount for the Program to Strengthen the Resilience of Vulnerable Agro-Pastoral Communities from 2020 to 2030 is 30 billion FCFA, (60MUSD) though no specific proportion is mapped out for gender related plans/activities.
- National Adaptation Plan (NAP) development: The UNFCCC adopted the Lima action plan in its process. This is a gender related plan that recognizes the full role women must play in fight against climate change to promote climate policies that promote gender equality and the integration of a gender perspective in the

NDCs implementation. Mali has developed several policies, strategies and action plans to checkmate climate change effect and its vagaries. These programs are being revised in order to develop the NAP by 2030 in order to better guide and coordinate actions. National Agriculture Policy and National Planning Policy are developed as part of the strategies to realize the NAP - promote sustainable, modern and competitive agriculture based primarily on family farms and give economic development planning a territorial dimension within the framework of an organization in the territory.

- Policy/legal frameworks: Measures to mitigate GHG emissions in the Agriculture sector focus on agriculture and livestock policies, plans and programs. The Mali Agricultural Development Policy objectives synchronize with the directives of the Agricultural Orientation Law taking into account the major challenges and issues facing Malian agriculture and reflect the country's vision. With strong ambitions to positively influence decision-making for the adoption of favorable policies for development of resilient climate change, Mali involves various levels of territorial collectivities (regions, District, circles and communes) in administrations and in mobilizing all local economic actors (citizens, companies, farmers, institutions) in managing people's efforts to fight against climate change. This is a commendable gender responsive approach.

SECTION 9. ACTION PLAN

Table 3. Climate change and gender action plan (Draft 1)

Objectives	Actions	Indicators	Implementing partners	Budget (USD)
Enhance the availability of land and improve access for women involved in farming	1.1. Identify regions where women's land access and ownership are low.	Number of regions	Minister Delegate to the Minister of Rural Development, responsible for Livestock and Fisheries, Ministry of Mines, Water and Energy, Ministry of Rural Development, NGOs	500,000
	1.2. Increase women's understanding of land ownership rights through sensitization outreach, particularly in those areas where women's land access and ownership are low.	Number of sensitizations	Ministry of Chiefs, Ministry of local government and Housing, Ministry of Agriculture, community radio stations and media	
	1.3. Involve traditional chiefs in land rights talks and awareness-raising initiatives		Chiefs, local government, Regional Directions of Agriculture	
	1.4. Create a media campaign and articles to raise men and women's understanding of their women's land rights, particularly through community radio in local languages.	Number of chiefs attending land right related meetings	Community radios, stations and media, National Extension Services, NGOs, Foundation and other private sector stakeholders	
		Number of awareness campaigns		

Foster the adoption of gender-responsive, climate-smart agricultural technologies, with a particular focus on women farmers.	2.1. Identify and prioritize climate-smart technologies that are responsive to gender considerations, including those related to water and nutrient management in rainfed and irrigated systems, varieties, mechanization and post-harvest suitable for use by women. 2.2. Disseminate information about the identified technologies, with a specific emphasis on business models for reaching out to women and women's organizations. 2.3. Provide women and women's groups with access to post-harvest technologies, such as parboilers, storage and food preservation methods. 2.4. Encourage women to use climate-smart agricultural equipment, such as solar powered irrigation systems, parboiling, direct seeding, mechanical transplanters, combined harvesters and similar mechanization innovations.	Number of climate smart technologies identified and prioritized Number of business models piloted Number of women and women groups accessing post-harvest technologies Number of climate smart equipment used by women	Minister Delegate to the Minister of Rural Development, responsible for Livestock and Fisheries, Ministry of Mines, Water and Energy, Ministry of Rural Development, NGOs Ministry of Chiefs, Ministry of local government and Housing, Ministry of Agriculture, community radio stations and media Chiefs, local government, Regional Directions of Agriculture Community radios, stations and media, National Extension Services, NGOs, Foundation and other private sector stakeholders	500,000

Enhance the capacity of extension services with a focus on gender and climate change considerations.	3.1.Set up community climate information centers to ensure access to climatic information for both urban and rural women farmers. 3.2.Enhance and establish new Farmer Training Centers (FTC) to improve agricultural training initiatives. 3.3.Develop and execute Training of Trainers (ToT) programs specifically designed for women involved in extension services cooperatives. 3.4.Equip existing agricultural extension officers with modern communication tools, such as iPads with weather apps, to enhance their capabilities.	Number of climate information centers established Number of farmer training centers established Number of trainers of trainers programs conducted Number of agricultural extension services equipped with modern tools	Minister Delegate to the Minister of Rural Development, responsible for Livestock and Fisheries, Ministry of Mines, Water and Energy, Ministry of Rural Development, NGOs and private sector	500,000
Enhance women farmers' access to climate-resilient agriculture by leveraging the current women's economic empowerment fund.	4.1.Establish a user-friendly fund specifically designed for promoting climate resilience among female farmers.	Number of women beneficiaries of the climate resilient funds	Minister Delegate to the Minister of Rural Development, responsible for Livestock and Fisheries, Ministry of Mines, Water and Energy, Ministry of	500,000

	4.2. Disseminate information about a novel funding opportunity tailored to support women farmers. 4.3. Acknowledge and promote effective approaches, such as rural seed banks, that contribute to climate resilience. 4.4. Encourage collaboration between rural banking and finance sectors to implement Climate-Smart Agriculture (CSA).	Number of women beneficiaries of the climate resilient funds Number of communication materials promoted Number of rural banks established Number of loans provided to rural women	Rural Development, NGOs and private sector	
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